

SHILPA REDDY

Data Engineer

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PROFILE SUMMARY

- Data Engineer with 5+ years of experience designing, developing, and managing end-to-end data pipelines and data lake solutions across AWS and Azure environments.
- Proficient in Big Data technologies such as Spark, PySpark, Hadoop, Hive, Sqoop, and Kafka for large-scale data processing and real-time streaming.
- Skilled in Python, Scala, Java, SQL, and PL/SQL for building automated ETL frameworks and integrating APIs to extract and transform diverse data sources.
- Experienced in using Databricks and Structured Streaming for distributed analytics and implementing medallion architecture for scalable, reliable data modeling.
- Skilled in automating data ingestion from diverse sources into cloud data lakes using ADF, AWS Glue, and Spark, enabling faster data availability for analytics and reducing latency in end-to-end pipelines.
- Hands-on experience with Snowflake, Redshift, and Azure Synapse for data warehousing, data modeling, and performance optimization.
- Adept in CI/CD automation using Jenkins, Terraform, and Azure DevOps, improving deployment consistency and release efficiency.
- Knowledgeable in data governance, security, and lineage with IAM, KMS, and Azure Purview for compliance and controlled access.
- Proficient in Docker and Kubernetes for containerized pipeline deployment and resource optimization.
- Collaborative team player with strong problem-solving skills, experienced in Agile/Scrum environments delivering reliable and business-impacting data solutions.

TECHNICAL SKILLS

Microsoft Azure Cloud	Azure Databricks, Azure data factory, Azure Synapse analytics/SQL (Data warehouse), Logic apps, Azure data lake, Azure Analysis services, Azure Key Vault services
Databases:	Snowflake, MySQL, Teradata, Oracle, MS SQL SERVER, PostgreSQL, DB2
Programming Languages:	Python, SQL, PL/SQL, Scala, Java, Shell Scripting
Big Data Ecosystem	Hadoop Map Reduce, Impala, HDFS, Hive, Pig, HBase, Flume, Storm, Sqoop, Oozie, Kafka, Apache Airflow, Spark, Zookeeper.
AWS	EC2, Amazon S3, ECS, Lambda, Glue, Kinesis, Step functions, Amazon RDS, VPC, IAM, Amazon Elastic Load Balancing, CloudFront, Auto Scaling, CloudWatch, Redshift
CI/CD & Containerization	Jenkins, Docker, Kubernetes
ETL Tools	Talend, Pentaho, Informatica, Ab Initio, SSIS
Version Control	Git, SVN, Bitbucket
Operating Systems	Windows, Windows, Linux, & Mac OS.
IDE Tools	Eclipse, PyCharm, Jupyter, Power BI, Tableau

EDUCATION

Indiana Wesleyan University, Indianapolis, USA

- Masters / Computer Science - Aug 2022 to May 2024

RELEVANT EXPERIENCE

Allstate Insurance, Northbrook, IL

Oct 2024 – Present

Data Engineer

- Designed and developed ETL workflows using Azure Data Factory and Databricks to ingest, transform, and load large policy, claims, and telematics datasets into the enterprise data lake.

- Built and optimized PySpark data processing pipelines to handle high-volume customer, agent, and policy transactions, ensuring data availability for analytics and underwriting systems.
- Architected a medallion-based Lakehouse on Azure Blob Storage and Delta Lake, standardizing bronze, silver, and gold layers to streamline reporting and predictive modeling.
- Integrated Kafka streaming to capture real-time driver telemetry and claims events, applying transformations in Spark Structured Streaming for downstream consumption.
- Automated data integration from on-premises databases such as MySQL into Azure Synapse and Snowflake using Azure Data Factory with dynamic pipelines, parameterized connections, and T-SQL scripts for post-load transformations and data validation.
- Leveraged Pandas within Databricks notebooks to perform advanced data cleansing, transformation, and validation of large policy and claims datasets, reducing downstream data errors by 25% and accelerating analytics readiness.
- Implemented Python and SQL scripts to apply business rules, validate data accuracy, and automate schema reconciliations across multiple insurance data domains.
- Created Power BI dashboards for business leaders to visualize customer retention metrics, risk categories, and claim resolution timelines.
- Designed star and snowflake schema models in Snowflake to support analytics and Power BI dashboards for risk profiling, policy renewals, and claim forecasting.
- Worked on Jenkins pipelines to orchestrate CI/CD processes using Git, Maven, and Bash scripting, executing unit, integration, and static analysis tests to ensure code quality and enable seamless deployment of new data pipeline releases across environments.
- Implemented Azure Logic Apps to orchestrate and automate end-to-end workflows such as ADF pipeline triggers, notification emails for failed jobs, and periodic data validation routines, enhancing operational efficiency and reducing manual interventions.
- Implemented data governance and security controls using Azure AD, Key Vault, and IAM policies to safeguard sensitive policyholder information and maintain compliance.
- Improved infrastructure automation scripts with Terraform for provisioning Databricks, ADF, and Synapse environments in a consistent and repeatable manner.
- Participated in Agile ceremonies and sprint reviews, working cross-functionally with data architects, business analysts, and QA teams to deliver high-quality data solutions aligned with business needs.

Environment: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Snowflake, Azure Blob Storage, Delta Lake, Kafka, PySpark, Pandas, Python, SQL, Power BI, Azure Logic Apps, Terraform, Git, Maven, Jenkins, Bash, Azure AD, Azure Key Vault, IAM, Docker, Kubernetes, Agile Methodology.

ACI Worldwide, Naples, Florida, USA

Jul 2023 - Sept 2024

Data Engineer

- Designed and implemented low-level Java APIs to efficiently ingest large volumes of structured and semi-structured data directly into HBase, enabling high-throughput data writes, optimized storage, and real-time accessibility for downstream analytics and processing.
- Developed ETL pipelines using Python and Snowflake to ingest, cleanse, and transform high-volume transactional and settlement data from multiple payment gateways into centralized data warehouses.
- Leveraged AWS Glue, Lambda, and Step Functions to automate batch and streaming data ingestion processes, reducing manual intervention and improving pipeline reliability.
- Built distributed Spark applications for large-scale data aggregation and reconciliation of authorization, clearing, and settlement records, improving fraud monitoring and reporting efficiency.
- Utilized Docker and Kubernetes to containerize and deploy Spark jobs, achieving fault tolerance and reducing system downtime during high-volume data loads.
- Executed SQL scripts and stored procedures, triggers, functions to handle incremental data loads, automate schema evolution, and perform consistency checks across merchant and transaction datasets.
- Migrated legacy on-prem data workloads to Snowflake, designing efficient data models for settlement, chargeback, and reconciliation workflows.
- Conducted query optimization and performance tuning in Snowflake using clustering, indexing, and query profiling, resulting in a 30% improvement in query response times.
- Provisioned and managed AWS EC2 instances and integrated Terraform for automated provisioning, improving environment scalability and consistency.
- Configured Jenkins pipelines for CI/CD to automate data workflow deployments and integrated static analysis checks to maintain code quality.
- Developed Kafka streams to capture near real-time transaction events for downstream fraud detection and reconciliation analytics.

- Created Power BI dashboards to visualize transaction volumes, settlement delays, fraud risk metrics, and merchant performance KPIs, improving business transparency and decision-making.
- Implemented monitoring solutions with CloudWatch and ELK Stack (Elasticsearch, Logstash, Kibana) to track data latency, processing errors, and system health.
- Used Scala and Spark SQL for developing custom data processing components to handle large, semi-structured financial event data efficiently.
- Collaborated with cross-functional business teams to translate transaction-level data into analytics-ready datasets for finance and compliance use cases.
- Participated in Agile ceremonies, conducted peer reviews, and contributed to production support by diagnosing pipeline failures and optimizing ETL performance under strict SLAs.

Environment: AWS Glue, AWS Lambda, AWS Step Functions, Amazon EMR, Amazon EC2, Amazon S3, Snowflake, HBase, Spark, Scala, Python, Docker, Kubernetes, Terraform, Kafka, Power BI, SQL, Jenkins, Git, ELK Stack (Elasticsearch, Logstash, Kibana), CloudWatch, Agile.

Dupont & IFF Merger, Hyderabad, India

Nov 2019 - Jul 2022

Data Engineer

- Developed analytical components using Scala, Spark, Apache Mesos, and Spark Streaming for real-time and batch data processing on Amazon EMR.
- Installed and configured Hadoop, MapReduce, Hive, and Pig on EMR clusters to handle large-scale data cleansing and pre-processing workflows.
- Managed and scheduled Spark cluster resources using Amazon EMR and containerized applications via Amazon EKS.
- Created cost optimization dashboards using Amazon QuickSight by analyzing AWS service usage and billing metrics.
- Built and maintained Spark-SQL-based data pipelines in Databricks integrated with S3 for transforming multi-format datasets (CSV, JSON, Parquet).
- Developed PySpark jobs to extract, transform, and aggregate customer usage data stored in S3, then loaded into Redshift for downstream analysis.
- Created Tableau dashboards after preparing and blending data using Alteryx and complex SQL queries from Oracle and SQL Server data sources.
- Built a reusable SFTP data ingestion framework using AWS Glue and Databricks to handle batch file transfers from external vendors.
- Monitored EMR, Glue, and Lambda jobs using Amazon CloudWatch to ensure data pipeline reliability and performance.
- Used Sqoop for importing and exporting structured data between on-prem Oracle databases and S3 via EMR.
- Deployed batch and streaming jobs in AWS Glue using Python for cleansing and transforming text and JSON data for analytics.
- Orchestrated workflows in Apache Airflow for scheduling Spark and Glue jobs, leveraging Python callable and branching operators.

Environment: AWS EMR, AWS Glue, AWS Lambda, Amazon Redshift, Amazon S3, Amazon EKS, Amazon QuickSight, Apache Airflow, CloudWatch, Python, Spark SQL, Scala, Hadoop, MapReduce, Hive, Pig, Databricks, Sqoop, Oracle, SQL Server, Alteryx, Tableau.